Internal Combustion Engine

It is a device or machine that converts the chemical energy of a fuel into heat energy by combustion of fuel and utilizes this heat energy to perform useful mechanical work.

Internal Combustion Engine: In which the combustion of fuel takes place inside the engine cylinder.

Examples: Petrol Engine, Diesel Engine

Classification of IC Engine:

i) According to number of cylinders

a) Single cylinder engine b) Multi cylinder engine: 2, 3,4,6,8 cylinder

ii) According to type of fuel used

a) Petrol Engine b) Diesel Engine c) Gas Engine d) Bio-fuel Engine

iii) According to method of ignition

a) Spark Ignition (SI) engine b) Compression Ignition (CI) engine

iv) According to number of stroke

a) Two stroke engine b) Four stroke engine

v) According to cycle of combustion

a) Otto cycle engine b) Diesel cycle Engine c) Duel Combustion cycle engine

vi) According to arrangement of cylinders

a) Inline Engine b) V-type engine c) Opposed type engine d) Radial Engine

vii) According to method cooling

a) Air cooled engine b) Water cooled engine

viii) According to use

a) Stationary engine b) Automobile engine c) Marine engine d) Aircraft engine

Components of I C Engines:

Cylinder: It is the heart of an I C Engine, as the name indicates is a cylindrical shaped component in which combustion of fuel takes place. The cylinder is usually made from gray cast iron or steel alloys to withstand the high pressure and temperature generated inside the cylinder due to combustion of fuel.

Piston: The piston is a close fitting hollow cylindrical plunger reciprocating inside the engine cylinder. The power developed by the combustion of the fuel is transmitted by the piston to the crankshaft through the connecting rod.

Piston Rings: Towards the top of the piston, a few grooves are provided to accommodate the piston rings.

The piston rings are two types: compression ring and oil ring

- Compression rings: These rings are press hard with the cylinder walls maintaining a tight seal between the piston and the cylinder. This is required to prevent the high-pressure gasses from escaping into the crankcase.
- Oil ring: The function of oil ring is to extract the excess lubricating oil from the cylinder walls and send it back to the oil sump through the holes provided on the piston.

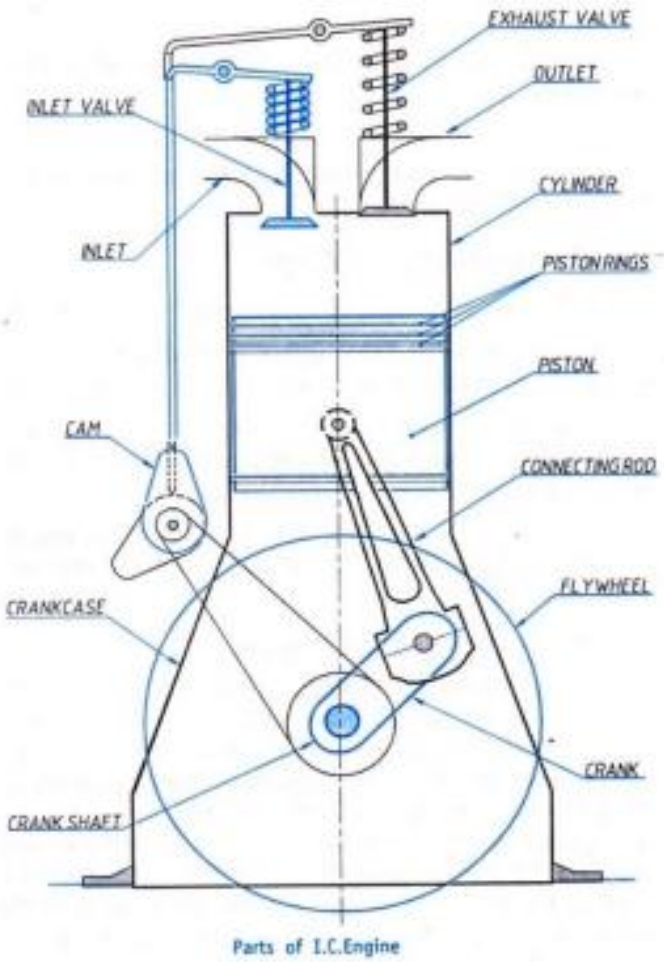


Figure: Components of IC engine

Connecting Rod: It connects the piston and the crankshaft. It converts the reciprocating motion of the piston into rotary motion of the crankshaft.

Crank: The crank is a lever, with one of its ends connected to the lower end of the connecting rod, while the other end connected to the crankshaft.

Crankshaft: The function of the crankshaft is to transform reciprocating motion into rotary motion. The crankshaft transmits the power developed by the engine through the flywheel, clutch, transmission and differential to drive (move) the vehicle. The crankshafts are made of carbon steel.

Crankcase: The crankcase is the lower part of the cylinder block that encloses the crankshaft and provides a reservoir for the lubricating oil.

Valves: The valves are the devices which controls the flow of the intake and the exhaust gases to and from the engine cylinder. They are also called poppet valves. These valves are operated by means of cams driven by the crankshaft through a timing gear or chain.

Flywheel: It is a heavy wheel mounted on the crankshaft of the engine to maintain uniform rotation of the crankshaft

Four Stoke Petrol Engine (S I Engine) OR Otto cycle engine:

The charge used in the four-stroke petrol engine is a mixture of Air and Petrol, which is supplied by carburetor in suitable proportion. The charge is ignited by the spark generated by a spark plug. Four stroke petrol engines are commonly used in motor bikes, cars etc.

The four different strokes are:

i)Suction stroke ii) Compression stroke iii) Power or Expansion stroke iv) Exhaust stroke

1. **Suction stroke**: During suction stroke the inlet valve opens, and exhaust valve is in closed condition. The piston starts moving from TDC to BDC, the volume above the piston increases, results in decrease in pressure (vacuum), this decrease in pressure draws the petrol and air mixture from the carburetor and delivered it to the cylinder, this process is continuous till the pressure inside the cylinder becomes equal to atmosphere. The energy required to perform this stroke is supplied by ‘cranking’ only during the first cycle at the time

of starting, while running, the flywheel supplies the mechanical energy. The suction stroke is completed by rotating the crankshaft from 0° to 180°.

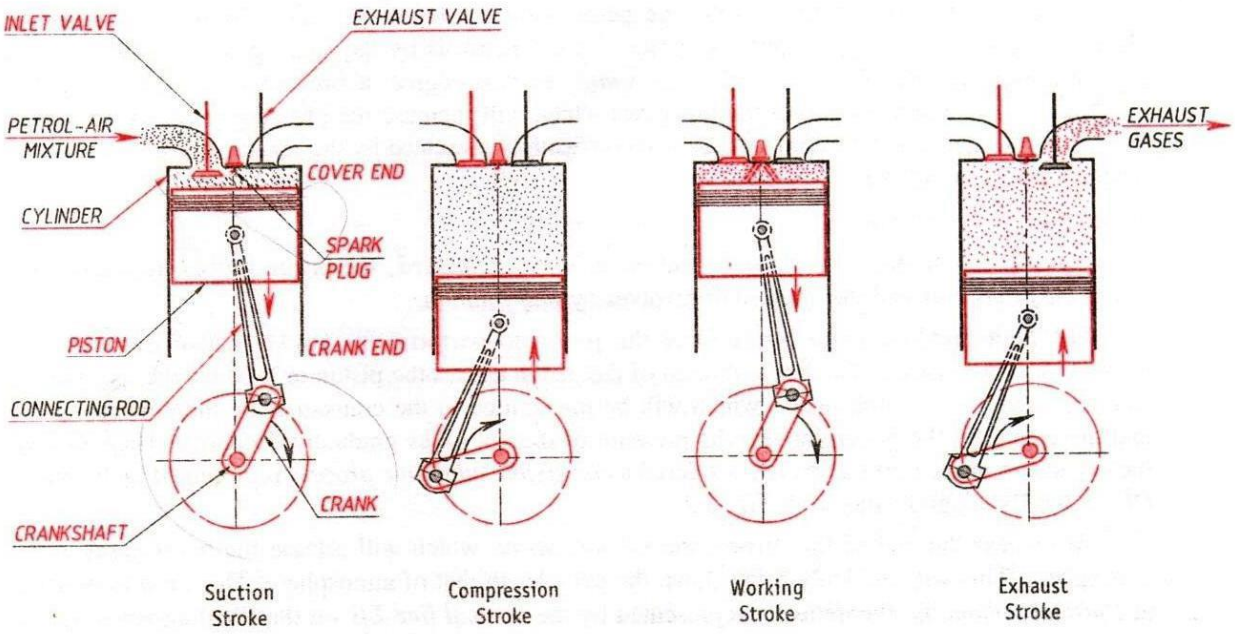
The line AB in the PV diagram represents suction stroke (volume of mixture filled in the cylinder).

2. **Compression stroke**: During the compression stroke both inlet and exhaust valves are closed and the piston travels from the BDC to TDC. The crankshaft revolves further by half rotation, pressure increases, volume decreases, causing the compression of air and fuel mixture.

   This stroke is represented by a line BC on the (Pressure -Volume) P-V diagram.

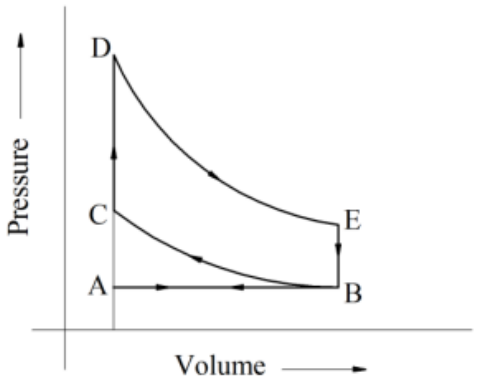
   At the end of this stroke a spark is produced by a sparkplug, resulting in the combustion of the fuel and air & is represented by a line CD on P-V diagram.

3. **Power stroke / Expansion stroke / Working stroke**: In this stroke the piston travels from TDC to BDC with both the valves remain closed & the crankshaft revolves half rotation. The piston is forced due to the expansion of the burnt gases. The linear motion of the piston is transmitted to the crankshaft through the connecting rod to produce Mechanical power. This stroke is called as power stroke as the Mechanical power is produced during this stroke. It is represented by the curve DE on a P-V diagram. As the piston moves further, the pressure of the hot gases gradually decreases at constant volume as represented by the line EB in PV diagram.



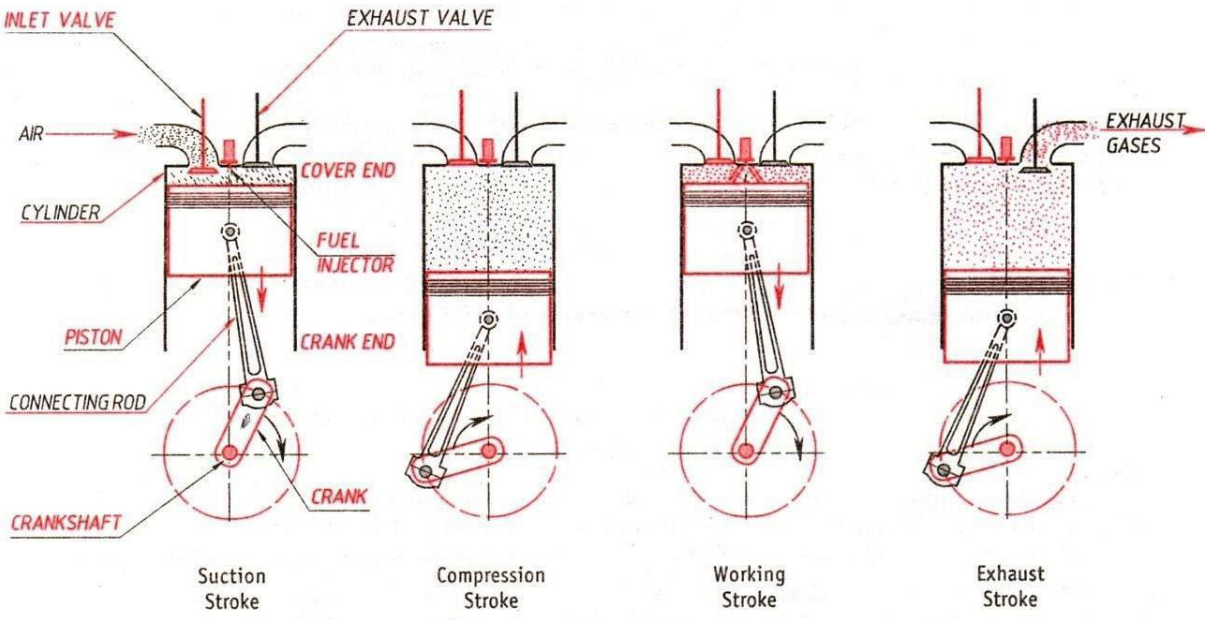
4. **Exhaust stroke:** During Exhaust stroke the exhaust valve opens with inlet valve closed and the piston travels from BDC to TDC, the crankshaft revolves half rotation. As the piston

moves upward the burnt gases are expelled out of the cylinder at atmospheric pressure. The energy required to perform this stroke is supplied by the flywheel from the energy absorbed by it during the previous stroke. This stroke is represented by a line BA on the (Pressure - Volume) P-V diagram.



4-Stroke Diesel Engine (Or constant pressure cycle Or Compression Ignition engines):
Since the ignition in these engines is due to high compression of air, it is very commonly known as Compression Ignition engines (C.I Engines). The four strokes that are performed are, 1. Suction stroke 2. Compression stroke 3. Working stroke or Power stroke or Expansion stroke & 4. Exhaust stroke

The following figure represent working principle of 4-stroke Diesel engine with Theoretical Diesel-cycle.



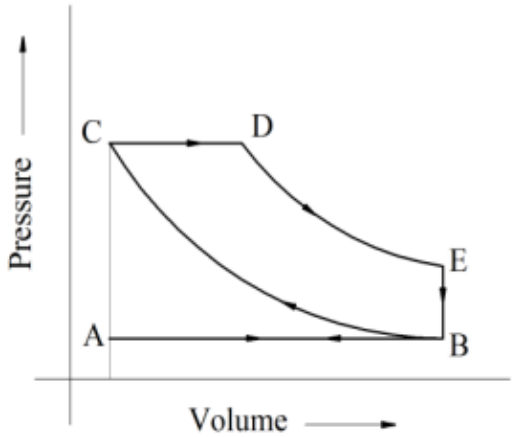
1. **Suction stroke:** During suction stroke the inlet valve opens with outlet valve closed & the piston travels from (Top Dead Center) TDC to (Bottom Dead Center) BDC & the crankshaft revolves by half rotation, the volume above the piston increases, results in decrease in pressure (vacuum), this decrease in pressure draws causing the suction of pure air. The energy required to perform this stroke is supplied by 'cranking' only during the first cycle at the time of starting, while running, the flywheel supplies the mechanical energy. This stroke is represented by a line AB on the (Pressure -Volume) P-V diagram.

2. **Compression stroke**: During the compression stroke both inlet and exhaust valves are closed and the piston travels from the BDC to TDC & the crankshaft revolves further by half rotation, causing the compression of air. This stroke is represented by a line BC on the (Pressure -Volume) P-V diagram. At the end of this stroke a metered quantity of fuel is injected through the fuel injector, the high temperature of the air ignites the fuel as soon as it is injected. This is called Auto-ignition or Self-ignition. This constant pressure expansion with simultaneous combustion is represented by the line CD on PV diagram.

3. **Power stroke / Expansion stroke / Working stroke:** In this stroke both inlet and exhaust valves remain closed & the crankshaft revolves half rotation. The burnt gases released by the combustion of the fuel that is continuously injected into the cylinder, forces the piston from TDC to BDC. This linear motion of the piston is transmitted to the crankshaft through the connecting rod to produce Mechanical power. This stroke is called as power stroke as the Mechanical power is produced during this stroke. It is represented by the curve DE on a P-V diagram.

Now the exhaust valve opens and the pressure of the hot gases gradually decreases at constant volume as represented by the line EB in PV diagram.

4. Exhaust stroke: During Exhaust stroke the exhaust valve remains open and inlet valve closed and the piston travels from BDC to TDC, causing the exhaust of burnt gases from the cylinder & the crankshaft revolves half rotation. This stroke is represented by a line BA on the (Pressure -Volume) P-V diagram.



Comparison between Petrol Engine & Diesel Engine:

Sl No	SI Engines (Petrol engine)	CI Engines (Diesel engine)
1	Ignition of the fuel by spark plug	Ignition of the fuel by spraying fuel to compressed air at high temperature
2	Works on theoretical Otto cycle	Works on theoretical Diesel cycle
3	Fuel used is petrol	Fuel used is diesel
4	A mixture of air and petrol is drawn during suction stroke	Only air is drawn during suction stroke
5	Combustion is at constant volume	Combustion is at constant pressure
6	Low compression ratio ranging from 7:1 to 12:1	High compression ratio ranging from 16:1 to 22:1
7	Fuel cost is high	Fuel cost is low
8	Power output will be less	Power output will be more

Following are the applications of IC engine:

- IC engines are used in Road vehicles like scooters, motorcycles, cars, buses etc.
- It is also used in Aircraft.
- IC engine is commonly used in Motorboats.
- IC engine has great application in small machines, such as lawnmowers, chainsaws, and portable engine-generators.
- IC engines are used in generators to produce electricity
- IC engines used in Diesel power plants

Important Questions:

1. Explain the Working of 4-Strokes Diesel Engine with P-V diagram.
2. Enumerate the Difference between Petrol engine and Diesel engine
3. What are the applications of IC engines?
4. Analyze the working of the 4 stroke Petrol engine with sketches. Plot the PV diagram.
5. List the differences between 4 stroke Petrol and diesel engine

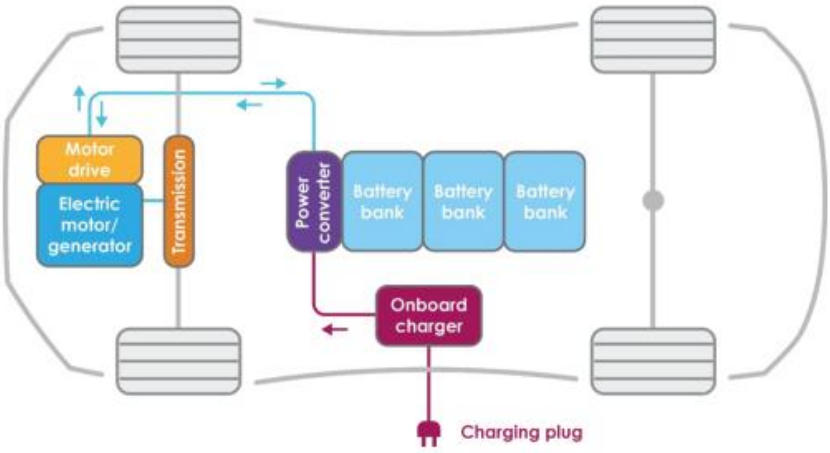
INSIGHT INTO FUTURE MOBILITY TECHNOLOGY

A 'EV' is defined as a vehicle that can be powered by an electric motor that draws electricity from a battery and is capable of being charged from an external source.
An EV includes both a vehicle that can only be powered by an electric motor that draws electricity from a battery (all-electric vehicle) and a vehicle that can be powered by an electric motor that draws electricity from a battery and by an internal combustion engine (**plug-in hybrid electric vehicle**).

- EVs are vehicles that are either partially or fully powered on electric power.
- Electric vehicles have low running costs as they have less moving parts for maintenance and also very environmentally friendly as they use little or no fossil fuels (petrol or diesel).

Components of Electric Vehicles:

1. **Traction Battery Pack**
   - The function of the battery in an electric car is as an electrical energy storage system in the form of direct-current electricity (DC).
   - The type of battery used is a rechargeable battery that is arranged in such a way as to form what is called a traction battery pack.
   - The most widely used is the type of lithium-ion batteries.



2. **DC/DC Converter**: This device converts higher-voltage DC power from the traction battery pack to the lower-voltage DC power needed to run vehicle accessories and recharge the auxiliary battery.
3. **Power Electronics Controller:** This unit manages the flow of electrical energy delivered by the traction battery, controlling the speed of the electric traction motor and the torque it produces.

4. **Charge Port**: The charge port allows the vehicle to connect to an external power supply in order to charge the traction battery pack.
5. **Electric Traction Motor**: Using power from the traction battery pack, this motor drives the vehicle's wheels. Some vehicles use motor generators that perform both the drive and regeneration functions.
6. **Onboard Charger**: Takes the incoming AC electricity supplied via the charge port and converts it to DC power for charging the traction battery.
7. **Thermal System (Cooling):** This system maintains a proper operating temperature range of the engine, electric motor, power electronics, and other components.
8. **Transmission (Electric)**: The transmission transfers mechanical power from the electric traction motor to drive the wheels.
9. **Power electronics controller:** It controls the flow of electrical energy from battery to motor and controls the speed.

Working Principle

Step I : Controller takes and regulates electrical energy from battery to inverter. converted from the DC battery to AC for the electric motor.
Step II : The inverter converts DC current from battery to AC then sends electrical energy to the motor.
Step III : The accelerator pedal sends a signal to the controller which adjusts the vehicle's speed by changing the frequency of the AC power from the inverter to the motor. The motor converts electrical energy into mechanical energy (rotation).
Step IV : Rotation of the motor rotor rotates the transmission, so the wheels turn and then the vehicle moves. Electric vehicles accelerate faster than the traditional fuel engines. So they feel lighter to drive.

ADVANTAGES OF ELECTRIC VEHICLES:

Electric vehicles (EVs) offer several advantages compared to traditional internal combustion engine (ICE) vehicles.

1. **Zero Emissions**: EVs produce zero tailpipe emissions, which means they do not release harmful pollutants like carbon dioxide (CO2), nitrogen oxides (NOx), particulate matter, and volatile organic compounds (VOCs). This significantly reduces air pollution and contributes to combating climate change.
2**. Reduced Greenhouse Gas Emissions**: EVs can be powered by electricity from renewable sources like solar, wind, or hydroelectric power, which can further reduce greenhouse gas emissions, making them more environmentally friendly.

3**. Energy Efficiency**: Electric motors are inherently more efficient than internal combustion engines. EVs convert a higher percentage of energy from the grid into usable power at the wheels, resulting in reduced energy waste compared to ICE vehicles.
4**. Lower Operating Costs**: Electric vehicles have fewer moving parts and require less maintenance than ICE vehicles. There are no oil changes, and brake wear is often reduced due to regenerative braking, which converts kinetic energy back into electricity. Over the vehicle's lifetime, this can lead to lower operating costs.
5. **Quiet and Smooth Operation**: EVs operate more quietly than traditional vehicles, which helps reduce noise pollution. They also provide smoother acceleration due to the instant torque provided by electric motors.
6. **Energy Independence**: EVs can be powered by locally generated electricity, reducing dependence on imported fossil fuels and enhancing energy security.
7. **Incentives and Rebates**: Many governments offer incentives, tax credits, and rebates to encourage the adoption of electric vehicles. These financial benefits can make EVs more affordable for consumers.
8. **Advancements in Battery Technology**: The continuous development of battery technology has led to improved energy density, longer range, and faster charging times for EVs, addressing some of the initial limitations of early electric vehicles.
9. **Health Benefits**: By reducing tailpipe emissions, EVs contribute to improved air quality, which has positive impacts on public health, reducing respiratory and cardiovascular illnesses associated with air pollution.
10. **Future Scalability and Grid Integration**: As renewable energy sources become more prevalent, EVs can play a vital role in energy storage and grid balancing. They can charge when there's excess renewable energy and discharge power back to the grid when needed, helping to create a more sustainable energy ecosystem.

DISADVANTAGES OF ELECTRIC VEHICLES

1. **Limited Driving Range**: Compared to traditional internal combustion engine (ICE) vehicles, EVs often have a limited driving range on a single charge. Although this has been improving with advancements in battery technology, some EVs may still not have the same range as their gasoline-powered counterparts.
2. **Charging Infrastructure**: The availability of charging infrastructure can vary significantly depending on the region. In some areas, charging stations may be scarce, which can cause "range anxiety" for EV owners, making long-distance travel less convenient.
3**. Charging Time**: Charging an EV takes longer than refueling a conventional car. Even with fast-charging options, it can take considerable time to recharge the battery fully. This can be a drawback for people with busy schedules or those who need to cover long distances regularly.
4. **Upfront Cost**: Electric vehicles tend to have a higher upfront cost compared to traditional ICE vehicles. Although this cost is gradually decreasing as battery technology advances, it remains a barrier for some potential buyers.

5**. Battery Life and Replacement Cost**: EV batteries have a limited lifespan, and over time, their capacity to hold a charge decreases. While advancements have been made to improve battery longevity, eventual replacement can still be a significant expense.
6. **Weight and Performance**: The batteries used in EVs can be heavy, affecting the overall weight and performance of the vehicle. While electric motors provide instant torque, some EVs may have reduced acceleration or handling capabilities due to their weight distribution.
7. **Dependency on Electricity Grid:** Charging an EV relies on access to electricity. In regions where the electricity grid is unreliable or experiences frequent power outages, charging an EV could be challenging.
8. **Environmental Impact of Battery Production**: The manufacturing process of batteries for EVs can have a significant environmental impact, including carbon emissions and resource extraction. Proper recycling and disposal of batteries are also critical to mitigate environmental concerns.
9. **Limited Model Variety**: While the availability of electric vehicle models is increasing, it may still be challenging to find specific types of EVs, such as larger SUVs or trucks, which might limit consumer choices.
10**. Perceived Performance and Driving Range**: Some consumers may still be hesitant to switch to EVs due to concerns about perceived performance, range, and the overall driving experience compared to traditional vehicles.

Hybrid vehicles

- It has both an Internal Combustion Engine (ICE) and an electric motor. In this type of electric cars, ICE gets energy from fuel while the motor gets electricity from batteries. It offers a mixture of battery and gasoline powers.

• A hybrid electric vehicle cannot be plugged in to charge the battery. Instead, the battery is charged through regenerative braking and by the internal combustion engine.

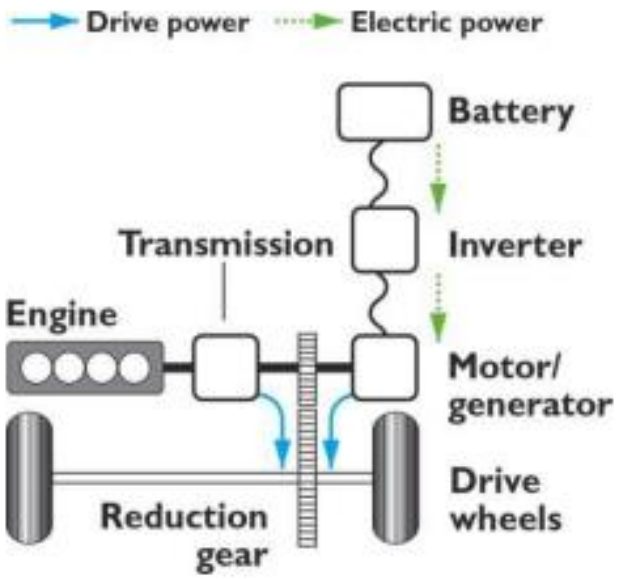
• The extra power provided by the electric motor can potentially allow for a smaller engine.

• The battery can also power auxiliary loads and reduce engine idling when stopped. Together, these features result in better fuel economy without sacrificing performance

Operation modes:

In a series hybrid system, the combustion engine drives an electric generator (usually a three-phase alternator plus rectifier) instead of directly driving the wheels. The electric motor is the only means of providing power to the wheels.
The generator both charges a battery and powers an electric motor that moves the vehicle. When large amounts of power are required, the motor draws electricity from both the batteries and the generator.

Series hybrid configurations already exist a long time: diesel-electric locomotives, hydraulic earth moving machines, diesel-electric power groups, loaders.

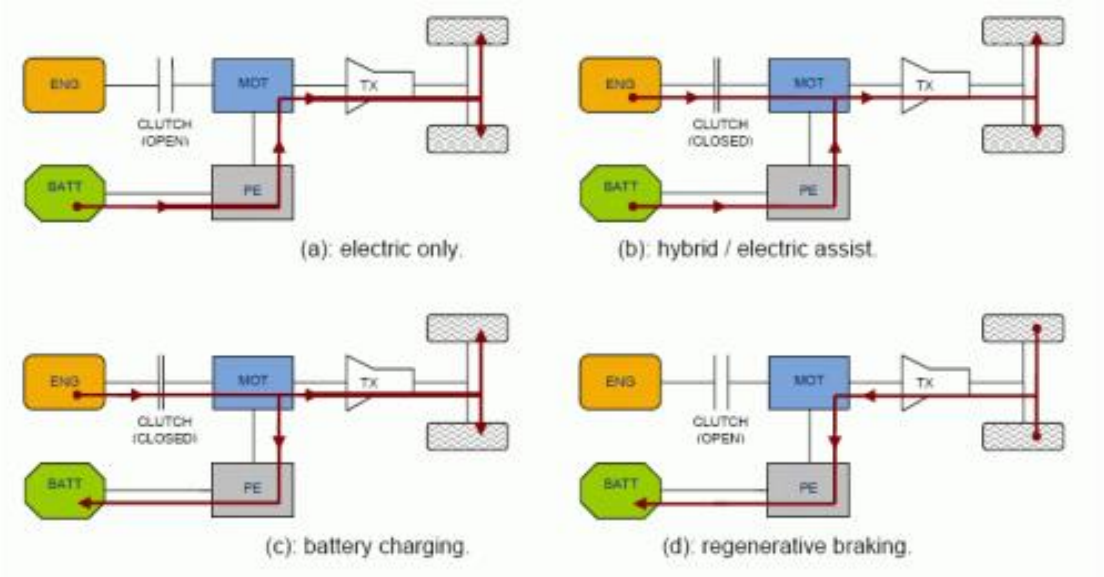


a) **electric power only**: Up to speeds of usually 40 km/h, the electric motor works with only the energy of the batteries, which are not recharged by the ICE. This is the usual way of operating around the city, as well as in reverse gear, since during reverse gear the speed is limited.

b**) ICE power only**: At speeds superior to 40 km/h, only the heat engine operates. This is the normal operating way at the road.

c) **ICE + electric power**: if more energy is needed (during acceleration or at high speed), the electric motor starts working in parallel to the heat engine, achieving greater power

d) **ICE + battery charging**: if less power is required, excess of energy is used to charge the batteries. Operating the engine at higher torque than necessary, it runs at a higher efficiency.



Components of Hybrid Vehicles:

1. **Battery (auxiliary):** In an electric drive vehicle, the low-voltage auxiliary battery provides electricity to start the car before the traction battery is engaged; it also powers vehicle accessories.
2. **DC/DC converter**: This device converts higher-voltage DC power from the traction battery pack to the lower-voltage DC power needed to run vehicle accessories and recharge the auxiliary battery.
3. **Electric generator**: Generates electricity from the rotating wheels while braking, transferring that energy back to the traction battery pack. Some vehicles use motor generators that perform both the drive and regeneration functions.
4. **Electric traction motor**: Using power from the traction battery pack, this motor drives the vehicle's wheels. Some vehicles use motor generators that perform both the drive and regeneration functions.
5. **Exhaust system**: The exhaust system channels the exhaust gases from the engine out through the tailpipe. A three-way catalyst is designed to reduce engine-out emissions within the exhaust system.
6. **Fuel tank (gasoline):** This tank stores gasoline on board the vehicle until it's needed by the engine
7. **Internal combustion engine (spark-ignited):** In this configuration, fuel is injected into either the intake manifold or the combustion chamber, where it is combined with air, and the air/fuel mixture is ignited by the spark from a spark plug.
8. **Power electronics controller:** This unit manages the flow of electrical energy delivered by the traction battery, controlling the speed of the electric traction motor and the torque it produces.
9. **Traction battery pack**: Stores electricity for use by the electric traction motor
10. **Thermal system (cooling):** This system maintains a proper operating temperature range of the engine, electric motor, power electronics, and other components.
11. **Transmission:** The transmission transfers mechanical power from the engine and/or electric traction motor to drive the wheels.

Advantages of Hybrid Vehicles:

- Environmentally Friendly
- Less Dependence on Fossil Fuels
- Regenerative Braking System
- Assistance From Electric Motor
- Smaller Engines

Disadvantages of Hybrid Vehicles:

- Less Power
- Can be Expensive
- Higher Maintenance Costs
- Battery Replacement is Pricey
- Battery Disposal and Recycling